

OAS-008 — Library Architecture

Status: Foundational

Authority: Architectural

Depends on:

- OAS-000
 - OAS-001
 - OAS-002
 - OAS-003
 - OAS-004
 - OAS-005
 - OAS-006
 - OAS-007
 - OCS-000 through OCS-032
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Purpose

This specification defines the architectural organization of Libraries.

The Constitution defines what a Library is.

This specification defines how Libraries are organized so that constitutional understanding remains discoverable, composable, and continuously expandable.

Libraries are not repositories.

Libraries are architectural knowledge organisms.

1. Architectural Objective

The objective of Library Architecture is to maximize the long-term discoverability, explainability, and reusability of constitutional understanding.

Architecture SHALL optimize for:

Discovery.

Connection.

Reconstruction.

Evolution.

Federation.

Not storage efficiency.

2. Architectural Organization

Libraries SHALL organize constitutional constructs through relationships.

Primary organizational dimensions MAY include:

Meaning.

Questions.

History.

Evidence.

Places.

Voices.

Patterns.

Dependencies.

Organization SHALL remain semantic.

Not physical.

3. Library Independence

Every Library SHALL remain architecturally sovereign.

Libraries MAY migrate.

Libraries MAY federate.

Libraries MAY evolve.

Architectural sovereignty SHALL remain preserved.

4. Library Membership

Architectural membership SHALL remain open.

Libraries MAY contain any constitutional construct.

The architecture SHALL remain extensible.

Future construct types SHALL integrate without architectural redesign.

5. Library Relationships

Relationships constitute the primary architectural structure.

Libraries SHALL organize:

Semantic relationships.

Historical relationships.

Methodological relationships.

Architectural relationships.

Constitutional relationships.

Relationship richness SHALL determine architectural quality.

6. Library Discovery

Architectural discovery SHALL occur through relationships.

Navigation SHALL prioritize:

Meaning.

Context.

Explanation.

Historical continuity.

Perspective diversity.

Discovery SHALL remain exploratory.

Not merely retrieval-oriented.

7. Library Composition

Libraries MAY compose into larger constitutional structures.

Composition SHALL preserve:

Local organization.

Independent history.

Semantic integrity.

Independent evolution.

Architectural composition SHALL remain recursive.

8. Library Federation

Independent Libraries MAY federate.

Federation SHALL preserve:

Local sovereignty.

Independent curation.

Independent history.

Shared constitutional meaning.

Architecture SHALL avoid centralized architectural dependency.

9. Library Evolution

Libraries SHALL evolve continuously.

Evolution SHALL preserve:

Canonical references.

Historical lineage.

Relationship continuity.

Semantic integrity.

Growth SHALL remain cumulative.

10. Library Curation

Architectural curation SHALL organize understanding.

Curation SHALL NOT redefine constitutional meaning.

Multiple curatorial perspectives MAY coexist.

Alternative navigational structures SHALL remain constitutionally compatible.

11. Library Reconstruction

Every Library SHALL remain reconstructable using:

Its constitutional identity.

Architectural relationships.

Historical lineage.

Canonical references.

Applicable constitutional specifications.

No implementation-specific organization SHALL be required.

12. Library Verification

Verification SHALL demonstrate:

Relationship preservation.

Semantic continuity.

Historical reconstruction.

Federation compatibility.

Independent discoverability.

Equivalent implementation SHALL NOT be required.

Equivalent architectural behavior SHALL.

13. Architectural Pattern

Libraries SHALL behave as evolving semantic ecosystems.

Growth SHALL increase navigability.

Growth SHALL increase connectivity.

Growth SHALL increase understanding.

Architecture SHALL preserve emergence rather than impose hierarchy.

14. Architectural Library Test

Library Architecture satisfies this specification when an independent architect can reconstruct:

How knowledge is organized.

How understanding becomes discoverable.

How historical continuity is preserved.

How federation occurs.

How curation evolves.

How future Libraries extend existing understanding.

If any answer cannot be reconstructed,

Library Architecture is incomplete.

Conformance

An implementation conforms to this specification when Libraries remain sovereign, semantically organized, historically continuous, architecturally reconstructable, and capable of indefinite constitutional evolution.

Architectural Principle

A repository stores knowledge.

A Library organizes knowledge.

A constitutional Library cultivates understanding.

Architecture therefore transforms collections into ecosystems,

allowing every discovery to become the beginning of future discoveries rather than the end of a search.

End of Specification OAS-008